

AMOGH SOMISETTY

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Seeking a Summer 2027 engineering internship

EDUCATION

California Polytechnic State University, San Luis Obispo

B.S. Electrical Engineering

Expected June 2028

San Luis Obispo, CA

- **Coursework:** Computer Architecture, Digital Logic Design, Embedded Systems, Signals & Systems, Electronics (in progress)

EXPERIENCE

Creator & Lead Researcher, Theta

Open-source GPU thermal-power forensics, advised by ex-Intel silicon-validation faculty

Mar 2026 – Present

San Luis Obispo, CA

- Blind-flagged 3 of **64 H100s** on **Princeton's** Della cluster from telemetry alone, two invisible to temperature thresholds; the operator independently confirmed two, one replaced under RMA and one re-measured at **+47.8%** thermal resistance.
- Shipped the agent behind it (`pip install runtheta`): peer-relative thermal resistance ($R_{\theta} = \Delta T/P$) by two-way median polish across position- and power-matched NVML/DCGM peers, flagging degradation in **6 minutes** against a multi-day ticket loop.
- Found the method's own failure mode on a rented 8xA100 node, where baseboard position alone moves healthy R_{θ} by **52%** at matched power, and rebuilt the detector position-conditioned to recover all 3 flags at zero false positives.
- Authoring two manuscripts for peer-reviewed publication: a five-fleet failure taxonomy across ~50,000 GPUs with a permutation null on every claim, and a detection-method paper.

Electrical Lead, Avionics & Power

Poly UAS, Cal Poly Unmanned Aerial Systems competition team

Apr 2026 – Present

San Luis Obispo, CA

- Own the aircraft's electrical system: designed a 48V and DroneCAN backplane in KiCad (2-layer, 120×90mm) that replaces a hand-wired harness on a 12S heavy-lift quad, **ERC and DRC clean**.
- Sized four independent 48V pass-throughs to **14A continuous and 30A burst** per IPC-2221 on 2oz copper, with dual-line CAN TVS, switchable 120Ω terminations, and a written bring-up procedure.
- Now designing the team's **NVIDIA Jetson AGX Orin** carrier board and sensor modules (schematic capture and layout, in progress) to move the aircraft from ground-controlled flight to onboard real-time computer vision.

Optimization Engineering Intern, promoted from Lab Operations Intern

Natera, clinical genetic testing

Jun 2025 – Sep 2025

Pleasanton, CA

- Cut validation and packaging cycle time from ~20 minutes to 10–12 (**over 40%**) on a high-throughput clinical line by mapping the workflow and eliminating redundant verification steps.

TECHNICAL PROJECTS

RISC-V CPU (RV32I) on FPGA

SystemVerilog, Vivado, Basys-3

- Implemented a multi-cycle full-ISA core from the RTL up (10-operation ALU, FSM control unit, 32×32 register file); closed timing through synthesis and place-and-route, tracing faults to the exact RTL by waveform debug.

Mixed-Signal DAC Signal Chain on a Custom PCB (with a lab partner)

AVR, LTspice, 2-layer OSH Park board

- Simulated, fabricated, and bench-characterized a binary-weighted 4-bit DAC driven by register-level firmware, measuring **62.5mV/step within ±1 LSB** across all 16 codes.

Pantry Shelf Vision (Justice Centered Edge AI)

YOLO11n, PyTorch, ONNX, Jetson Orin Nano

- Built a 3-stage detection pipeline (SKU-110K pretrain, gap fine-tune, 15-frame on-site fine-tune) reaching **0.920 mAP50**, eliminating every false positive from a documented retail-to-site domain gap.
- Live-tested the deployed monitoring loop against a real camera before hardware bring-up, finding and fixing **4 defects**, including a monitor that reported total blindness as “all shelves OK.”

TECHNICAL SKILLS

Hardware & EDA KiCad (schematic capture, multilayer layout, fabrication), LTspice, IPC-2221 conductor sizing, power delivery and protection, board bring-up, oscilloscope, DMM, SMD/reflow soldering

Digital & Embedded SystemVerilog RTL, Vivado (synthesis, place & route, static timing), FPGA bring-up (Artix-7), self-checking testbenches, waveform debug, AVR and STM32 register-level firmware, CAN/DroneCAN, I2C, SPI, UART

Software & Systems Python, C, Go, Bash, SQL, NVML, DCGM, Prometheus, SLURM, Kubernetes (Helm, client-go), Docker, Linux, Git, GitHub Actions CI, pytest, NumPy, pandas, scikit-learn

LEADERSHIP

President, Poly Engineering Consulting: lead client-facing engineering project teams

Sep 2024 – Present

Lab Assistant, Cal Poly EPIC: taught embedded systems to precollegiate students

Jun 2026 – Aug 2026